

we are not quite there yet, and still have a long trek ahead. A crucial reason for that is the FOSS ecosystem. "Though FOSS in India is growing at a faster rate than in the past, the FOSS ecosystem will take time to mature, as awareness of the potential of FOSS is yet to cross the threshold level. Training and certification bodies have grown, and [so has] the use of FOSS on servers—but the user base in India is still not large enough," says J A Bhavsar, CIO, ITM Group.

So what's hindering the development of a strong and supportive FOSS ecosystem? "A FOSS ecosystem consists of numerous stake-holders—educational institutions like schools and colleges, government, FOSS solution providers, consumers and the FOSS community. Though we have large volumes of people in India who are now aware of FOSS, we remain largely consumers. Poor community participation, lack of proper revenue/business model, lack of well established governance model in leveraging FOSS, very few India based ISVs have all hindered the eco system. Creating awareness is just the first step, and resolves only one dimension of building the ecosystem. We still have a long way to go," says Jyothi Bacche, head, Open Source Practice, MindTree.

Stakeholders in the FOSS ecosystem

Participants in the Indian FOSS ecosystem fall into three broad categories:

- Producers, including developers and those who are contributing back to FOSS projects. Examples are the JBoss India chapter and the MySQL Indian community.
- Consumers—basically those that use FOSS, including SMEs, large enterprises, and government.
- Facilitators, like academia and FOSS solution providers, who connect producers and consumers, and build a demand-and-supply chain around FOSS.

All these stake-holders need to play their role for the ecosystem to mature, and for India to become a FOSS hub.

Producers—critically required

Every nation has its own indigenous technology-related

Talent—a major bottle-neck within the ecosystem

The lack of a large talent pool is a big hindrance for companies that use FOSS. Speaking based on his experience in embedded systems, Sridharan Mani, director and CEO, American Megatrends, says, "The industry is seeking the best embedded talent, to develop and deliver solutions to clients the world over. But acquiring suitable candidates and training them with the desired skill-sets is a huge effort, involving time, resources and investment. To make India an 'open embedded' hub, we need to start at the basics—our education system. For instance, our curriculum for embedded engineering programs should encourage students to learn through examples; innovations should be recognised to promote original thinking; more weight should be given to practicals; and there should be a platform for students to come together, showcase their talents, compete and brainstorm—probably something on the lines of the Science Olympiad."

To address these talent-related issues, FOSS vendors are providing the much-needed challenging and learning environment for developers to grow. For instance, Aspire Systems encourages its employees to actively participate in community forums and contribute back to the community. Their 'Wisdom curve' sessions help employees learn new open source tools and keep abreast with the latest open source skillsets. Similarly, American Megatrends (AMI) is working on building a talent pool that is well-equipped with FOSS skills for innovation and research. "We have been recognising such talent over the years. We provide fresh engineers in embedded technology an opportunity to live through real-life projects in a simulated environment, through our AMI Challenge (an internal competition). We also encourage employees to share their knowledge through embedded technology presentations, white-papers and case studies," says Mani.

Zynga India also does its bit to motivate the Indian FOSS community. "We do hackathons, code jams, etc, wherein multiple developers can come in and contribute. Further, our hiring strategy is open source friendly; about 90 per cent of our team comes from the open source world creating a large demand for FOSS skill sets," says Kadavil. EnterpriseDB employs seven key PostgreSQL contributors, including Pavan Deolasee, who is a major contributor to the development of HOT (Heap-Only-Tuples), one of the most important performance improvement features in PostgreSQL 8.3.



"Companies that use FOSS to build products ought to credit FOSS when advertising their products; this can help create a healthy awareness and prove the utility and widespread application of FOSS."

Rohit Kumar Rai, sales director, India, SE Asia and Australia, EnterpriseDB